Advanced robust design method for optical coatings

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ABSTRACT

The development of high-performance optical coatings demands advanced methods that reconcile the simultaneous optimization of multiple competing performance criteria—including spectrum, phase, group delay dispersion, and LIDT, with the imperative of ensuring manufacturability. Broad Band Optical Monitoring, a key high-precision in-situ monitoring technique, often introduces correlated thickness errors during deposition, directly compromising spectral performance and repeatability. To address these challenges, we present a robust design method specifically for correlated BBOM errors, ensuring spectral stability under manufacturing variations. The core of this method lies in utilizing an analytical model of production errors to predict the actual production yield of a coating design. By explicitly optimizing the design to maximize this predicted yield, the method ensures spectral stability under realistic manufacturing variations. Its effectiveness is validated through computational simulations. The results conclusively show that these methods significantly enhance the comprehensive performance and production yield of optical coatings.

Keywords: Design method, optical monitoring, robustness, muti-objective

1. INTRODUCTION

Optical coatings play a crucial role in modern industrial applications, serving as key components for achieving high-performance optical devices. Their precise design is essential, as it directly impacts the overall performance and efficiency of these devices [1]. With the increasing demand for high-performance optical components, design objectives have expanded beyond conventional spectral responses to include phase, dispersion, optical loss, and laser-induced damage threshold (LIDT) [2]. These advanced requirements impose stringent constraints on fabrication accuracy, making the control of manufacturing errors a critical challenge. Consequently, ensuring manufacturability, particularly the robustness against manufacturing errors, has become an increasingly important consideration [3].

Robust design methods aim to enhance the optical performance of optical coatings against manufacturing errors. The primary goal of robust design is to minimize the sensitivity of the final optical performance to random and systematic variations in layer thicknesses and refractive indices, which are the most common manufacturing deviations. By incorporating statistical information about these errors into the design phase, robust optimization seeks to produce coatings with a high production yield—meaning a large percentage of fabricated samples will meet the required specifications. This shifts the focus from finding a single, theoretically perfect design to identifying a design that performs optimally across a range of probable manufacturing outcomes. Existing robust design approaches mainly include methods like analytical approximation and Monte Carlo simulation. The first involves analytical approximation methods, which use a linear expansion of the merit function around the target design to estimate its sensitivity to small parameter variations. The second, more computationally intensive but accurate approach is the Monte Carlo method. Here, a large number of virtual coatings are generated by randomly perturbing the design parameters according to a predefined statistical distribution. The performance of these virtual samples is then statistically analyzed, typically by calculating the mathematical expectation of the merit function or directly estimating the production yield. These methods effectively improve coatings robustness by calculating the mathematical expectation of the merit function or production yield. However, they typically assume that layer-wise manufacturing errors are independent, an assumption that may not hold under advanced deposition

In modern fabrication processes, to achieve the high precision required for advanced optical coatings, deposition processes increasingly rely on optical monitoring techniques, which offer the distinct advantage of compensating for

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manufacturing errors in real-time during deposition. Among these, broadband optical monitoring (BBOM) has become particularly prevalent due to its high accuracy and information richness. However, this compensation capability introduces a fundamental challenge: it creates correlated errors. Unlike non-optical monitoring methods where thickness errors in each layer are statistically independent, BBOM determines each layer's thickness by fitting the transmission or reflection spectrum of the entire growing stack. This causes errors to accumulate and propagate through subsequent layers in a complex, correlated manner. Present robust design methods, which primarily focus on non-correlated errors from non-optical monitoring, are therefore inadequate for analyzing and simulating these correlated error structures. This leaves a critical gap in robust design methodologies for modern optical coatings.

In this paper, a robust design method for correlated errors in broad band optical monitoring is proposed, and its robustness is validated through a series of computational manufacturing simulations.

2. ROBUST DESIGN METHOD FOR BBOM

In the robust design of optical coatings fabricated with broadband optical monitoring (BBOM), accurate error analysis is essential, as the optimization objective must explicitly account for the impact of manufacturing uncertainties on the final spectral performance. In particular, the correlated and cumulative nature of BBOM-induced errors makes their statistical characterization a critical prerequisite for robust optimization. Computational manufacturing provides a rigorous approach for such error analysis by numerically simulating the full deposition process, including signal generation and spectral fitting. Through this process, both the thickness errors—along with their self-compensation effects—and the resulting spectral deviations can be accurately obtained. However, due to its multi-stage iterative structure, computational manufacturing is computationally expensive and therefore unsuitable for robust design, where repeated evaluations are required during optimization. To overcome this limitation, a fast analytical method [4] is developed based on the underlying error propagation mechanisms in BBOM. This method enables efficient estimation of the statistical properties of correlated manufacturing errors and their impact on the optical spectrum.

The calculation method for each set of correlated thickness errors is as follows

$$\delta d_J = \sum_{I=1}^{J} \beta_I^J \Delta_p,$$
$$\begin{cases} \beta_I^J = \alpha_I^J + \sum_{K=1}^{J-I-1} \alpha_{I+K}^J \beta_I^{I+K}, \alpha_I^J = -\left(\sum_{\lambda} \frac{\partial T_J}{\partial d_I} \frac{\partial T_J}{\partial d_J} \right) \Big/ \sum_{\lambda} \left(\frac{\partial T_J}{\partial d_J} \right)^2, I < J \\ \beta_I^J = 1, I = J, \end{cases} \tag{1}$$

Where, δd_J is the accumulated thickness error, Δ represents the random errors occurs in each layer, including those lead by shutter delay and spectrum errors. It can be written as

$$\Delta = \delta d_{shutter} + \delta d_{spectrum} = v t_{shutter} - \sum_{\lambda} \frac{\partial T_i}{\partial d_i} \delta T(\lambda) \Big/ \sum_{\lambda} \left(\frac{\partial T_i}{\partial d_i} \right)^2,$$
$$\delta T(\lambda) = \delta T_{meas}(\lambda) + \sum_{q=1}^{j} \frac{\partial T_q(\lambda)}{\partial n_q(\lambda)} \delta n_q(\lambda) \tag{2}$$

Where, $t_{shutter}$ is time of shutter delay, δn_q is the error of refractive index, δT_{meas} is the errors of measured spectrum.

Based on this formulation, a Taylor expansion of the spectral response with respect to these perturbations is performed. Combined with statistical analysis, analytical expressions for the expectation and standard deviation of the spectrum are obtained:

$$E(\Delta T) = \sum_{i=1}^{m} \frac{\partial T^0}{\partial d_i} E(\delta d_i) + \frac{1}{2} \sum_{i,j=1}^{m} \frac{\partial^2 T^0}{\partial d_i \partial d_j} E(\delta d_i \delta d_j) \tag{3}$$

$$D(\Delta T) = \sum_{i,j=1}^{m} \frac{\partial T^0}{\partial d_j} \frac{\partial T^0}{\partial d_i} Cov(\delta d_i, \delta d_j) + \sum_{i,j,k=1}^{m} \frac{\partial T^0}{\partial d_i} \frac{\partial^2 T^0}{\partial d_j \partial d_k} Cov(\delta d_i, \delta d_j \delta d_k)$$
$$+ \sum_{i,j,r,s=1}^{m} \frac{\partial^2 T^0}{\partial d_i \partial d_j} \frac{\partial^2 T^0}{\partial d_r \partial d_s} Cov(\delta d_i \delta d_j, \delta d_r \delta d_s). \tag{4}$$

With these statistical quantities, the effect of correlated errors can be efficiently evaluated without resorting to computational manufacturing. Based on the above analytical results, a merit function for robust design is constructed as:

$$MF_{robust} = \sqrt{\frac{1}{N}\sum_{\lambda}\left(\frac{E \pm 2D - T}{\Delta}\right)^2} \tag{5}$$

Where, E and D are the mathematical expectation and standard deviation of the actual spectrum considering the influence of errors, calculated according to Equation 4 and Equation 5, respectively, T is the target spectrum, and Δ is the spectrum tolerance. The inclusion of the standard deviation term effectively characterizes the spectral envelope under uncertainty (approximately corresponding to a 95% confidence interval). By minimizing this merit function, the spectral variation range is reduced, thereby improving robustness against correlated manufacturing errors and increasing production yield.

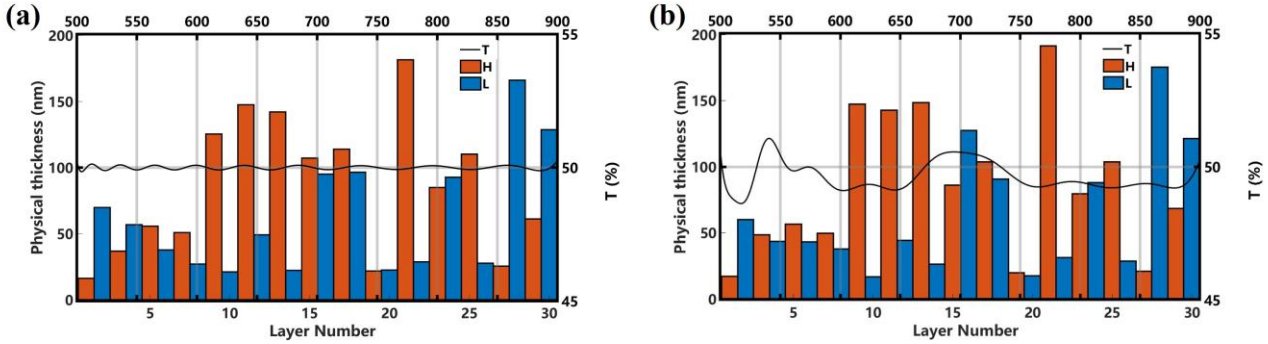


Figure 1. The thickness and theoretical transmittance of the 50/50 beam-splitting coatings optimized by (a) conventional method, (b) robust method.

We designed two broadband dielectric broadband beamsplitters for the 500–900 nm wavelength range at an incidence angle of 22.5°, based on the conventional method and the proposed robust method, respectively. The incident medium and substrate are air and Glass7980.Ta2O5 and SiO2 are chosen as the high and low refractive index materials respectively. The design target was to achieve equal reflectance and transmittance (R=T=50%) over the entire wavelength range under averaged polarization. Fig. 1 show the physical thickness distributions and the theoretical spectral performance of the two designs, respectively. The theoretical results indicate that both designs achieve near-ideal splitting performance, with the conventional design slightly outperforming the robust design in theory, while the subsequent computational manufacturing results demonstrate that the robust design achieves superior actual performance.

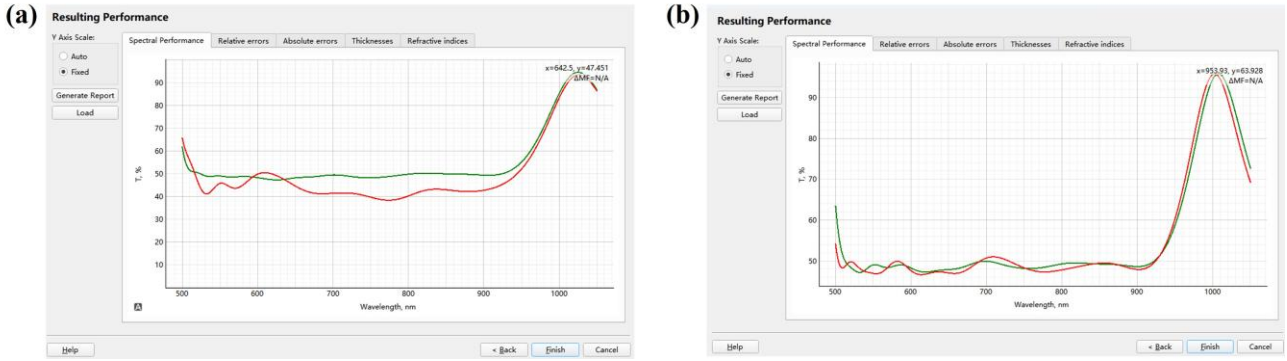


Figure 2. Computational manufacturing simulation results for the (a) conventional design and the (b) robust design.

To evaluate the practical manufacturing performance, 100 BBOM-based computational simulations were carried out for each design. The simulation parameters were set as follows: monitoring at a 0° incident angle over a wavelength range of 400–1100 nm, spectral random error with a standard deviation of 0.5%, shutter delay with a standard deviation of 1 s, and a deposition rate of 0.4 nm/s with a standard deviation of 0.04 nm/s. Fig. 2 presents one of the simulated spectra. As shown in the figure, the robust design achieves a splitting ratio closer to the target value across the entire wavelength range compared to the conventional design. Statistical analysis based on the 100 simulations reveals that for the conventional design, the transmittance fluctuates significantly beyond the 45–55% range, with particularly pronounced deviations in the

650–900 nm region. In contrast, the robust design consistently maintains the transmittance within the 47.5–52.5% range throughout the entire spectrum. Its spectral envelope becomes significantly narrower, indicating substantially reduced sensitivity to manufacturing errors. These results demonstrate that the proposed method effectively suppresses variations in splitting performance under correlated manufacturing errors, leading to significantly enhanced robustness.

3. CONCLUSION

A robust design method for optical coatings fabricated with BBOM is presented, addressing correlated manufacturing errors. Using Taylor expansion, we derived analytical expressions for spectral expectation and standard deviation under correlated errors. A merit function incorporating both terms was constructed to minimize performance variability. Computational simulations of 50/50 beam-splitting coatings demonstrated that while the conventional design showed marginally superior theoretical performance, the robust design achieved better practical results with significantly narrower spectral envelopes, confirming reduced sensitivity to manufacturing errors. This method bridges the gap in robust design for BBOM-fabricated coatings.

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Use this section to identify people who have aided the authors in accomplishing the work presented and to acknowledge sources of funding. Include grant numbers and the full name of the funding body.

REFERENCES

[1] A. V. Tikhonravov, and M. K. Trubetskov, "Modern design tools and a new paradigm in optical coating design," Appl. Opt. 51, 7319-7332 (2012).
[2] J. Dai, et al., "Efficient multi-objective design method for optical coatings." Optics and Lasers in Engineering 184, 108626 (2025).
[3] J. Dai, et al., "Efficient multi-objective design method for optical coatings." Optics and Lasers in Engineering 184, 108626 (2025)
[4] J. Dai, et al., "Error perturbation analysis method in broad band monitoring of optical coating," Optics & Laser Technology **163**, 109333 (2023).